Thresholding Operators for Complex Images

1 Summary

Problem formulation.

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}\in\mathbb{C}^N} \ \tfrac{1}{2}\|\mathbf{x}-\mathbf{z}\|_2^2 + \lambda R(\mathbf{x})$$

- $\mathbf{z} \in \mathbb{C}^N$: observed complex image.
- $\mathbf{x} \in \mathbb{C}^N$: estimate.
- $R(\mathbf{x})$: sparsity-promoting penalty, $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_0$ or $\|\mathbf{x}\|_{1/2}^{1/2}$.
- $\lambda > 0$: weight between the data term and the penalty; real, since the objective must be real-valued to be minimized.

Separability. If $R(\mathbf{x}) = \sum_i R_i(x_i)$,

$$\begin{aligned}\min_{\mathbf{x}\in\mathbb{C}^N} \ \tfrac{1}{2}\|\mathbf{x}-\mathbf{z}\|_2^2 + \lambda R(\mathbf{x}) &= \min_{\mathbf{x}\in\mathbb{C}^N} \ \sum_{i=1}^{N}\Big[\tfrac{1}{2}|x_i - z_i|^2 + \lambda R_i(x_i)\Big] \\ &= \sum_{i=1}^{N} \min_{x_i\in\mathbb{C}} \ \Big[\tfrac{1}{2}|x_i - z_i|^2 + \lambda R_i(x_i)\Big],\end{aligned}$$

so each $\hat{x}_i$ is found independently. Below, the index i is dropped ($x = x_i$, $z = z_i$, $R = R_i$):

$$\hat{x} = \arg\min_{x\in\mathbb{C}} \ \tfrac{1}{2}|x-z|^2 + \lambda R(x).$$

Phase. If $R(x) = r(|x|)$ (true for $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_0$, $\|\mathbf{x}\|_{1/2}^{1/2}$, defined through $|x_i|$), then for fixed $|x|$ the penalty is constant and $|x-z|^2 \geq (|x|-|z|)^2$, with equality iff x has the phase of z:

- Polar, $x = \rho e^{j\theta}$, $z = m e^{j\varphi}$: $|x-z|^2 = \rho^2 + m^2 - 2\rho m\cos(\theta-\varphi) \geq (\rho - m)^2$, "=" iff $\theta = \varphi$.
- Rectangular, $x = a + jb$, $z = c + jd$: $|x-z|^2 = |x|^2 + |z|^2 - 2(ac+bd) \geq (|x|-|z|)^2$, since $ac + bd \leq |x||z|$ (Cauchy–Schwarz), "=" iff $(a,b) = \kappa(c,d)$, $\kappa \geq 0$.
- Triangle inequality: $|x-z| \geq \big||x|-|z|\big|$, "=" iff $x = \kappa z$, $\kappa \geq 0$.

Hence, with $\rho = |x|$ and $m = |z|$,

$$g(\rho) = \tfrac{1}{2}\big(|x|-|z|\big)^2 + \lambda r(|x|) = \tfrac{1}{2}(\rho - m)^2 + \lambda r(\rho), \qquad \hat{\rho} = \arg\min_{\rho\geq 0} \ g(\rho), \qquad \hat{x} = \operatorname{sgn}(z)\,\hat{\rho}.$$

ℓ_1: soft thresholding. $r(\rho) = \rho$, $g(\rho) = \tfrac{1}{2}(\rho - m)^2 + \lambda\rho$, $g'(\rho) = \rho - m + \lambda$:

- $m > \lambda$: $g'(\rho) = 0 \Rightarrow \hat{\rho} = m - \lambda$.
- $m \leq \lambda$: $g'(\rho) > 0$ for $\rho > 0 \Rightarrow \hat{\rho} = 0$.

$$\mathcal{S}_\lambda(z) = \operatorname{sgn}(z)\ \max(|z| - \lambda,\, 0).$$

ℓ_0: hard thresholding. $r(\rho) = \mathbf{1}[\rho > 0]$, $g(\rho) = \tfrac{1}{2}(\rho - m)^2 + \lambda\,\mathbf{1}[\rho > 0]$:

- $\rho = 0$: $g(0) = \tfrac{1}{2}m^2$.
- $\rho > 0$: best $\rho = m$, $g(m) = \lambda$.
- $\hat{\rho} = m$ iff $\lambda < \tfrac{1}{2}m^2$, i.e. $m > \sqrt{2\lambda}$ ($\hat{\rho} = 0$ at equality).

$$\mathcal{H}_{\sqrt{2\lambda}}(z) = z \cdot \mathbf{1}\big[|z| > \sqrt{2\lambda}\big].$$

2 $\ell_{1/2}$: Half Thresholding (To Be Organized)

$R(\mathbf{x}) = \|\mathbf{x}\|_{1/2}^{1/2} = \sum_i |x_i|^{1/2}$ satisfies the Separability and Phase conditions of Section 1 with $r(\rho) = \sqrt{\rho}$, and $g(\rho) = \frac{1}{2}(\rho - m)^2 + \lambda\sqrt{\rho}$. Let $\mu = 2\lambda$ and minimize $2g(\rho) = (\rho - m)^2 + \mu\sqrt{\rho}$.

Stationary Point. For $\rho > 0$, $\frac{d}{d\rho}\big[(\rho - m)^2 + \mu\sqrt{\rho}\big] = 2(\rho - m) + \frac{\mu}{2\sqrt{\rho}}$. With $u = \sqrt{\rho}$, setting it to zero gives

$$u^3 - m\,u + \tfrac{\mu}{4} = 0. \tag{1}$$

Its largest root, the local minimum of g, is $u = 2\sqrt{m/3}\,\cos(\psi/3)$ with $\psi = \arccos\big(-\frac{\mu}{8}(\frac{m}{3})^{-3/2}\big)$. With $\phi = \pi - \psi$,

$$\rho^* = u^2 = \frac{2m}{3}\Big[1 + \cos\Big(\frac{2\pi}{3} - \frac{2}{3}\phi\Big)\Big], \qquad \phi = \arccos\Big(\frac{\mu}{8}\Big(\frac{m}{3}\Big)^{-3/2}\Big). \tag{2}$$

Cut-Off. ρ^* beats $\rho = 0$ iff $(\rho^* - m)^2 + \mu\sqrt{\rho^*} < m^2$. At the cut-off $m = t$, equality holds together with (1). Substituting $m = u^2 + \mu/(4u)$ from (1):

$$\big(\tfrac{\mu}{4u}\big)^2 + \mu u = \big(u^2 + \tfrac{\mu}{4u}\big)^2 \iff u^3 = \tfrac{\mu}{2},$$

so

$$t = u^2 + \frac{\mu}{4u} = \frac{3}{2}\Big(\frac{\mu}{2}\Big)^{2/3} = \frac{\sqrt[3]{54}}{4}\,(2\lambda)^{2/3}, \tag{3}$$

and at the cut-off $\rho^* = (\mu/2)^{2/3} = \frac{2}{3}t$.

Result.

$$\mathcal{T}_\lambda(z) = \begin{cases} \operatorname{sgn}(z)\,\dfrac{2|z|}{3}\Big[1 + \cos\Big(\dfrac{2\pi}{3} - \dfrac{2}{3}\phi(|z|)\Big)\Big], & |z| > t, \\ 0, & |z| \le t, \end{cases} \qquad \phi(m) = \arccos\Big(\frac{\lambda}{4}\Big(\frac{m}{3}\Big)^{-3/2}\Big). \tag{4}$$

The output jumps from 0 to $\frac{2}{3}t$ at the cut-off, and the shrinkage $m - \rho^* \approx \lambda/(2\sqrt{m})$ vanishes for large m.

3 Comparison of Thresholding Operators (To Be Organized)

Each operator keeps the phase and thresholds the magnitude, $\hat{x} = \operatorname{sgn}(z)\, f(|z|)$:

	cut-off t on $\lvert z\rvert$	$\lvert\hat{x}\rvert$ for $\lvert z\rvert > t$	jump at t
ℓ_1	λ	$\lvert z\rvert - \lambda$	none
ℓ_0	$\sqrt{2\lambda}$	$\lvert z\rvert$	$0 \to t$
$\ell_{1/2}$	$\frac{\sqrt[3]{54}}{4}(2\lambda)^{2/3}$	Eq. (2)	$0 \to \frac{2}{3}t$